

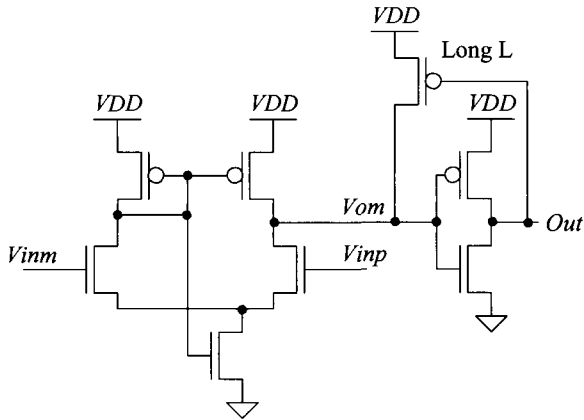


Figure 18.25 An input buffer for high-speed digital design with hysteresis.

18.3.3 DC Reference

In the previous section we assumed we had a DC voltage, V_{inm} , at precisely the correct voltage to slice the data (in the middle). In a real system the data varies and the communication channel is bandlimited (behaves like a lowpass filter). The result is that the amplitudes of the data vary depending on the data and the channel frequency response. Consider the simple RC lowpass filter and data used to model a transmission system as seen in Fig. 18.26. The ideal point where we slice the output data changes with the input data. What we need is a circuit that determines the maximum and minimum of an input waveform and outputs the average of the two to slice the buffer's input data in the center.

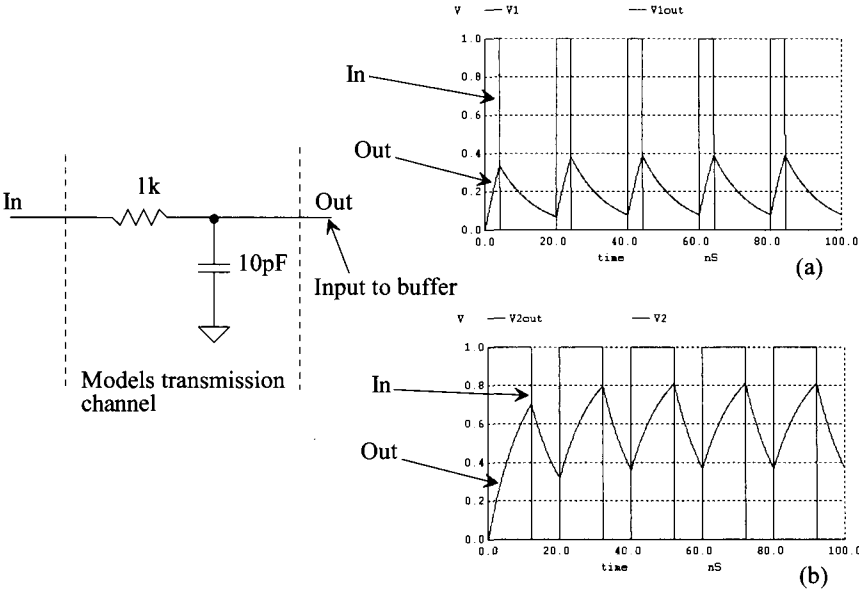


Figure 18.26 How the shape of the data changes through a communication channel.

Towards this goal consider the peak detector circuit seen in Fig. 18.27. The buffer from Fig. 18.23 can be used in this circuit. When the input, v_{in} , goes above the voltage stored on the capacitor, v_{peak} , the output of the buffer goes low, turning on the long-length MOSFET and pulling the output towards VDD . As v_{peak} approaches v_{in} , the MOSFET starts to shut off. As a result the output voltage across the capacitor, v_{peak} , corresponds to the peak voltage of the input signal. This peak detector can be used to generate a reference voltage that falls within the middle of the input data, Fig. 18.28. The peak and valley detectors are used to find the minimum and maximum of the input signal. The two resistors average the minimum voltages and feed the result (the DC average of the input) to the bottom buffer circuit. The resistors also are used to leak charge off of the capacitors so that the averaging circuit can follow changes in the input data (the averaging is actually a *running average*). Figure 18.29 shows some simulation results. In the top plots the output signal from Fig. 18.26a is applied to the input of the DC generation circuit. As expected, the output of the DC generation circuit, after a start-up delay, goes right to the middle of the input data.

The speed (response time) of this generation circuit can be increased by reducing the resistors and capacitors in the circuit. However, if the values are reduced too much, a long string of zeroes or ones will cause the DC generation circuit's output (labeled "average" in Fig. 18.28) to go to ground or VDD . The bottom traces in Fig. 18.29 show the results of applying the output signal from Fig. 18.26b to the DC generation circuit. Again, the output moves quickly to the center of the data. What's more interesting in these figures is a comparison between the original input data and the regenerated output data (labeled *In* and *Out* in Fig. 18.29). The output data pulse widths are considerably different from the input pulse widths (remembering that there is a delay through both the channel RC seen in Fig. 18.26 and the buffer in Fig. 18.28). Ideally, the pulse widths (the data) of the input and output are the same. Looking at the responses in Fig. 18.26a, we see that the effect of the finite channel bandwidth is to distort the channel's output data. *The only solution to this problem is to make the channel transmission bandwidth effectively wider.* An equalizer (discussed in the next chapter) can be used for this purpose at the cost of reduced signal swing. We can also reduce the input impedance of the input buffer to lower the time constant associated with the channel. Let's discuss this second approach.

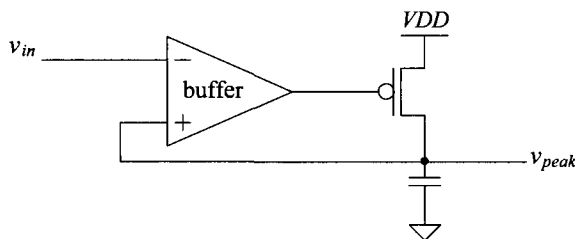


Figure 18.27 CMOS peak detector.

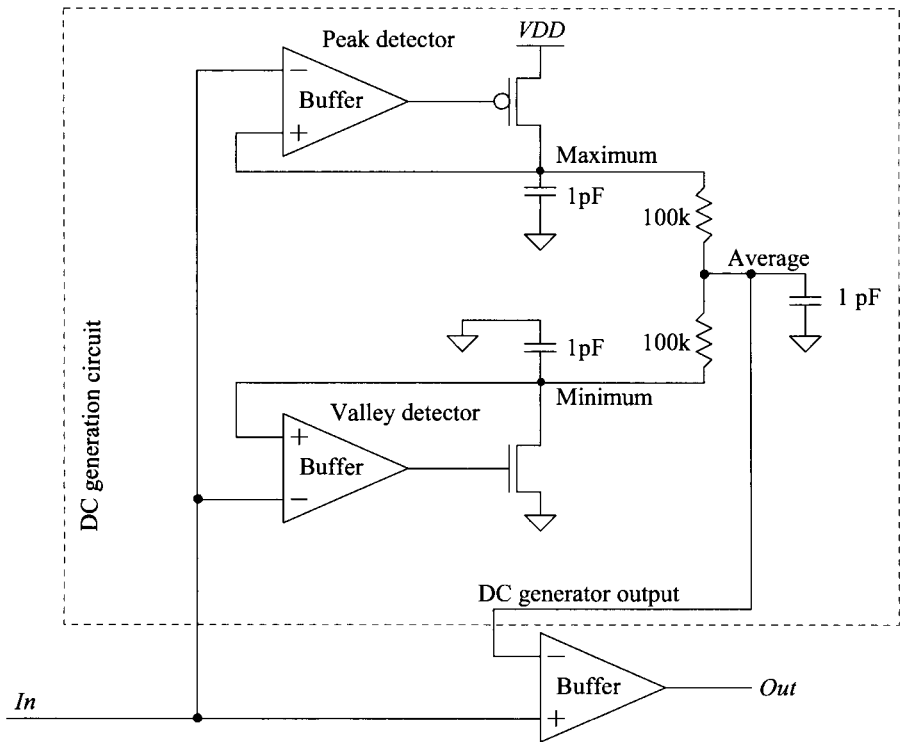


Figure 18.28 A DC generation circuit.

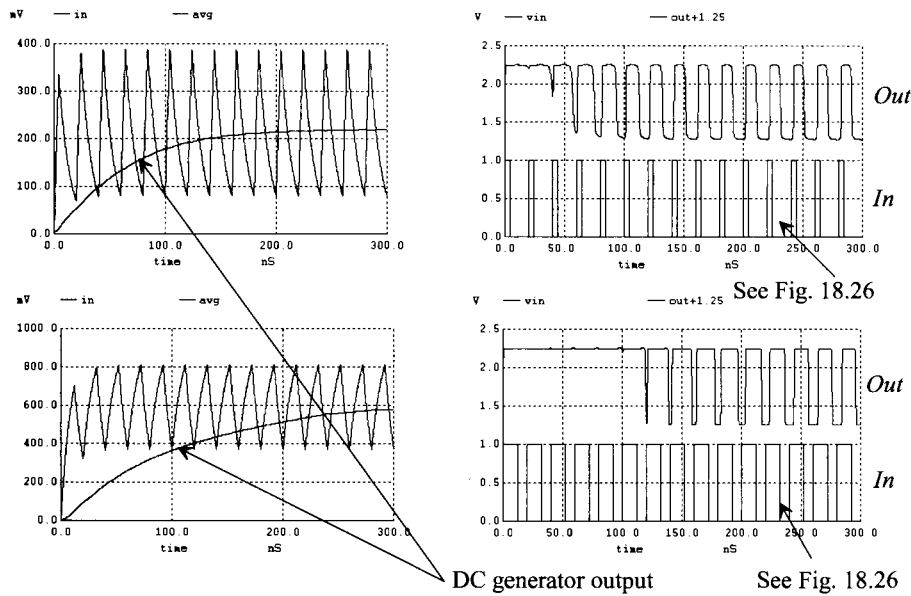


Figure 18.29 The output of the DC restore circuit in Fig. 18.28.

18.3.4 Reducing Buffer Input Resistance

A simple solution to increasing the effective bandwidth of the transmission channel is to reduce the input resistance of the input buffer. The time constant associated with the channel in Fig. 18.30 is 10 ns. When we connect the buffer to the transmission channel, the time constant drops to 3.33 ns (the three 1k resistors in parallel multiplied by the 10 pF capacitance). The drawbacks of this approach are increased power dissipation and reduced input signal amplitudes. Figure 18.31 shows how the buffer in Fig. 18.30 behaves with the input signals seen in Fig. 18.26a. The distortion (difference in the pulse widths of ones and zeroes when comparing the communication channel input, *In*, and the buffer output, *Out*) is much better.

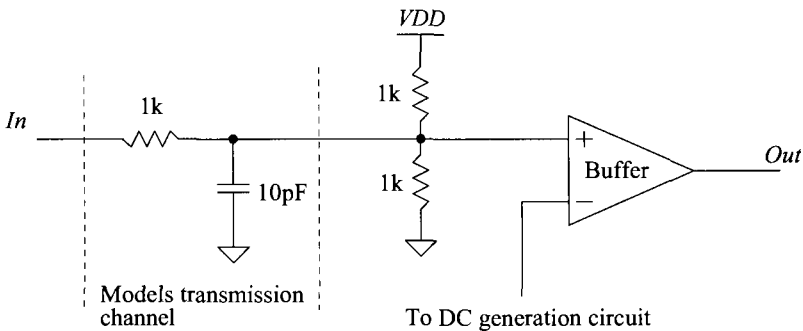


Figure 18.30 Reducing the input resistance of the input buffer.

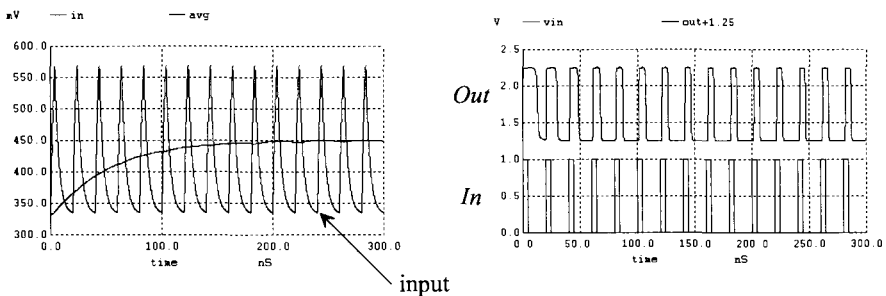


Figure 18.31 How reducing the input resistance of the buffer reduces the input signal amplitude (bad) and distortion (good).

Figure 18.32 shows an alternative low input resistance input buffer. When the input goes high, M1 turns on and M4 shuts off. This causes M2 to turn on and M5 to shut off. The output then goes high. When the input goes low, M1 turns off and M4 turns on (with M2 shutting off and M5 turning on). The result: the output goes low. Again, the power burned by M1 and M4 can be high if the transistors aren't sized properly. Further, the input swing is limited to a threshold voltage away from the supply voltages. This reduces the input voltage swing and further enhances the speed.

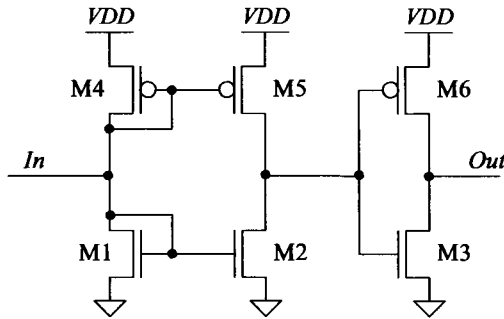


Figure 18.32 A low-input resistance input buffer.

18.4 Charge Pumps (Voltage Generators)

Often, when designing CMOS circuits, positive and negative DC voltages are needed that do not lie between ground and V_{DD} . An example of an application where a larger DC voltage source is needed was seen in Ch. 16 when we discussed Flash memory (see Fig. 16.59). A simple circuit, sometimes called a voltage pump (or more often, a *charge pump*), useful in generating a voltage greater than V_{DD} , is seen in Fig. 18.33.

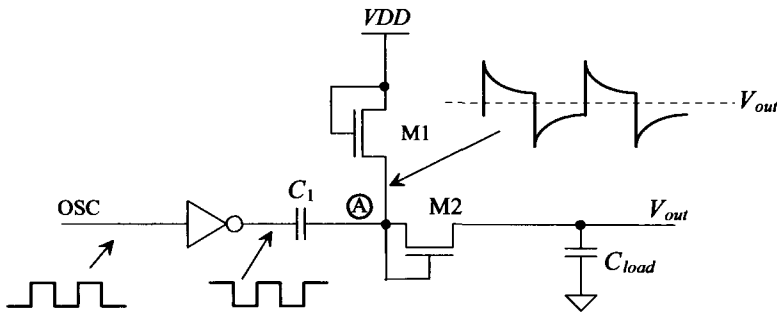


Figure 18.33 Pump used to generate a voltage greater than V_{DD} .

The operation of the voltage pump of Fig. 18.33 can be explained by first realizing that both M1 and M2 operate like a diode. M1 is simply used to pull point A to a voltage of $V_{DD} - V_{THN}$. M2 allows the charge from C_1 to charge C_{load} but not vice-versa. Let's begin the description of the circuit operation by assuming that the output of the inverter is low and point A is at a potential of $V_{DD} - V_{THN}$. When the output of the inverter goes high, the potential at point A increases to $V_{DD} + (V_{DD} - V_{THN}) = 2 \cdot V_{DD} - V_{THN}$. This turns on M2 and charges C_{load} to $2 \cdot (V_{DD} - V_{THN})$, [an extra V_{THN} because of M2's gate-source voltage drop], provided $C_1 \gg C_{load}$ and the oscillator frequency allows the capacitors to fully charge or discharge before changing states. In most practical situations, C_{load} and C_1 are comparable in size, and the output of the pump is loaded with a

DC load. The result is an output voltage with a startup time; that is, V_{out} does not immediately rise to $2 \cdot (V_{DD} - V_{THN})$ but requires several oscillator cycles to reach steady state. Also, the output has a ripple dependent on the DC load. Figure 18.34 shows the simulation results for the circuit of Fig. 18.13 using 10/1 NMOS (scale factor of 50 nm) with $C_{load} = C_1 = 1$ pF and an oscillator frequency of 10 MHz. Using this simple pump with a DC load, such as a resistor, may require employing larger MOSFETs and capacitors to avoid excessive voltage droop.

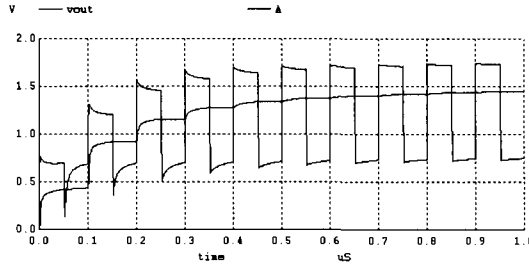


Figure 18.34 Simulating the operation of the voltage pump in Fig. 18.33.

Negative Voltages

The positive voltage pump uses n-channel MOSFETs, while the negative voltage pump, Fig. 18.35, uses p-channel MOSFETs. The reason for this comes from the requirement that the diode formed with the n+ (p+) implant used in the drain/source of the MOSFET combined with the p-substrate (n-well) does not become forward-biased. Forward biasing this parasitic diode is an *important concern* and is often the reason why some more exotic pump topologies can't be implemented. For example, in Fig. 18.35, we connect the well of the PMOS devices to ground instead of to their respective sources. Consider what would have happened had we connected the well to the source of the MOSFET. When the output voltage goes negative, the (n-type) well goes negative too. If the substrate (p-type) is at ground, this forward biases the n-well to substrate diode acting to clamp the output of the pump at a negative diode drop. While we could have left the bodies of the PMOS (the n-wells) tied to V_{DD} , here we chose to connect them to ground so that the body effect wasn't so severe (a lower threshold voltage).

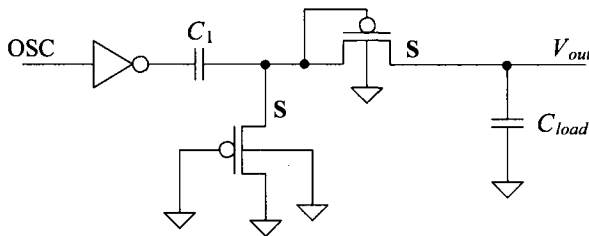


Figure 18.35 Negative voltage pump.

Using MOSFETs for the Capacitors

The capacitors used in Figs. 18.33 and 18.35 can be replaced with MOSFETs. An example of using n-channel MOSFETs in place of the capacitors in Fig. 18.33 is shown in Fig. 18.36. The main requirement on a MOSFET used as a capacitor is that its V_{GS} remain greater than V_{THN} at all possible operating conditions. In other words, the MOSFET must remain in the strong inversion region so that its capacitance is a constant $C'_{ox} \cdot W \cdot L$. The capacitor, C_{load} , in Fig. 18.36 remains in strong inversion because $V_{out} \gg V_{THN}$. The capacitor C_1 remains in strong inversion because, when the inverter output is low, the voltage on the gate of C_1 is $VDD - V_{THN}$ ($= V_{GS}$). When the output of the inverter is high (VDD), the voltage on the gate of C_1 is $2 \cdot VDD - V_{THN}$. In both cases (the output of the inverter high and low), $V_{GS} = VDD - V_{THN}$, and the MOSFET is in the strong inversion region.

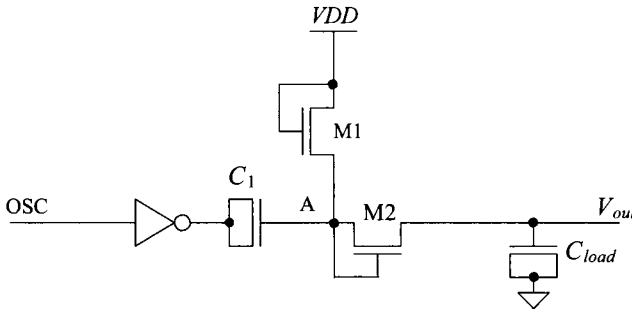


Figure 18.36 Using MOSFETs as capacitors.

18.4.1 Increasing the Output Voltage

Figure 18.37 shows a voltage pump with a higher output voltage. The increase in the output voltage comes from eliminating the threshold voltage drop at the gate and drain of M7. This allows the output to swing up to $2 \cdot VDD - V_{THN}$.

To understand the operation of the circuit, let's assume that the voltage at point A is low and the voltage at point C is $VDD - V_{THN}$. When the output of INV1 goes high, point A is VDD and the voltage at point C swings up to $2 \cdot VDD - V_{THN}$. This causes M4, M5, and M6 to turn on and pull points D and E to VDD . Now when point B goes high (point A goes back to zero), points D and E swing up to $2 \cdot VDD$ and the output goes to $2 \cdot VDD - V_{THN}$. Note that MOSFETs M2 and M3 are not needed; unless the pump drives a DC load, they never turn on. Also, separating points D and E is unnecessary unless the pump supplies a DC current.

18.4.2 Generating Higher Voltages: The Dickson Charge Pump

The voltage pumps of the last section are limited to voltages less than $2 \cdot VDD$ and greater than $-2 \cdot VDD$. Figure 18.38 shows a scheme for generating arbitrarily high voltages (limited by the breakdown voltage of the capacitors or the oxide breakdown of the MOSFETs). Again, the MOSFETs are used as diodes in this configuration.

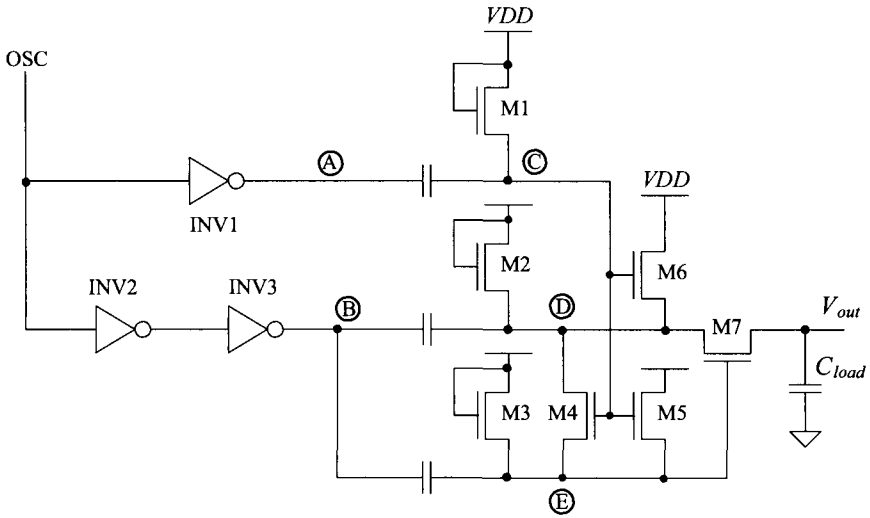


Figure 18.37 Increased output voltage pump.

In the following description of circuit operation, we assume steady-state operation with no DC load present. When CLK is low, point A is pulled, by M1, to $V_{DD} - V_{THN}$. When CLK goes high, point A swings up to $2 \cdot V_{DD} - V_{THN}$. Diode M2 turns on, and point B charges to $2 \cdot V_{DD} - 2 \cdot V_{THN}$. When CLK goes low, $\overline{\text{CLK}}$ goes high and point B swings up to $3 \cdot V_{DD} - 2 \cdot V_{THN}$. When $\overline{\text{CLK}}$ goes low, point B swings back down to $2 \cdot V_{DD} - 2 \cdot V_{THN}$. This operation proceeds through the circuit to the last stage of the multiplier. The output of the multiplier swings from $(N+1) \cdot V_{DD} - N \cdot V_{THN}$ down to $N \cdot V_{DD} - N \cdot V_{THN}$. C_N can be made larger than the other capacitors to reduce this ripple.

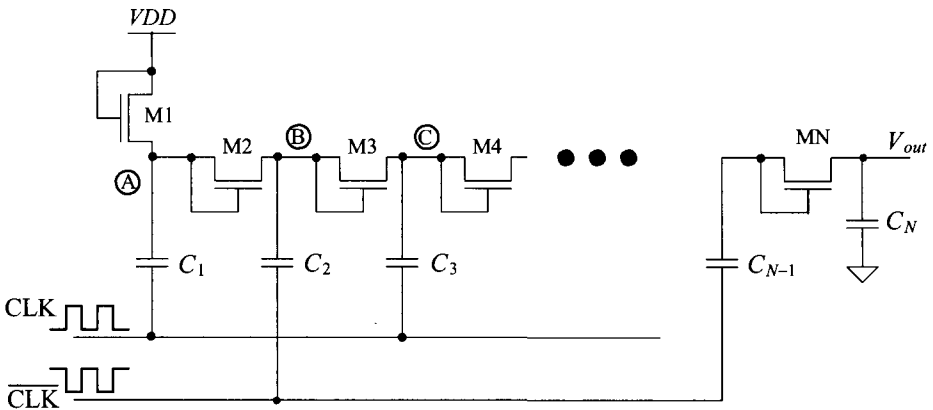


Figure 18.38 N-stage voltage multiplier (the Dickson charge pump).

Clock Driver with a Pumped Output Voltage

To fully turn on NMOS switches, a driver configuration is needed with a pulsed output voltage greater than $V_{DD} + V_{THN}$ (with body effect). One such configuration is seen in Fig. 18.39. When the input to the circuit is a logic low, the output of INV1 is a high and the output of INV2 is a low. Node B in the figure is at roughly V_{DD} , and node A is at roughly $2V_{DD}$. M1 is off and M2 is on. The output of the driver is a logic low (ground). When the input of the circuit goes high, the output of INV1 goes low. This causes node A to go to V_{DD} . The output of INV2 goes high, and node B boots up to $2V_{DD}$ (M1 on M2 off). The output of the driver circuit then goes to $2V_{DD}$ as well. Notice that one side of the cross-coupled NMOS devices is sized smaller than the other side. This is to minimize layout area and power (the charging and discharging of the parasitic capacitors doesn't waste power). Node A doesn't supply any power to the output of the circuit. It is simply used to turn M2 on so that node B is precharged to V_{DD} when the input to the circuit is low. A larger capacitor is used on node B because the charge that goes to load must be supplied by this capacitor. If the output load capacitance gets large, then the size of this capacitor must be increased. When designing the driver, it's a good idea to characterize the performance as a function of the load capacitance.

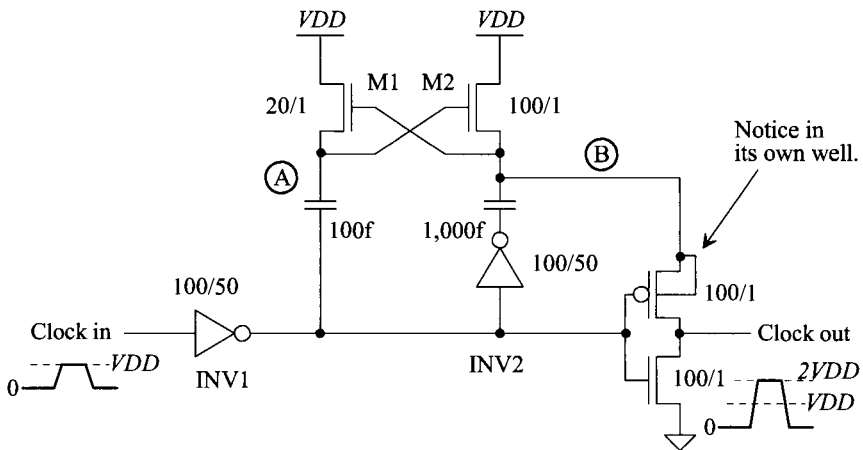
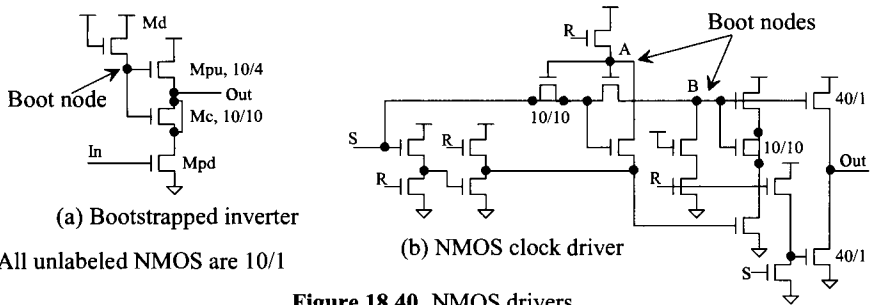


Figure 18.39 Charge-pump clock driver.

NMOS Clock Driver

Figure 18.40a shows an NMOS bootstrapped inverter. This circuit is useful to avoid latch-up since an n-well isn't present. The pull-up device, Mpu, is made 4 times longer than the pull-down device, Mpd, so that the output voltage can swing closer to ground. The transistor, Mc, is used as a capacitor to "boot" the gate of Mpu above V_{DD} so that it can turn fully on and drive the output to V_{DD} . Md is used as a diode, as in the other charge pumps seen in this chapter, to ensure that the boot node doesn't drop below $V_{DD} - V_{THN}$. Note that this isn't a static circuit, that is, the input must be active else the high output voltage is only $V_{DD} - 2V_{THN}$. In order to avoid continuous current flow when both Mpd and Mpu are conducting the NMOS "clock" driver (a buffer) seen in Fig. 18.40b can be used. The 40/1 output devices are both actively driven.



18.4.3 Example

A common application of a voltage generator in digital circuits is generating a negative substrate bias; that is, instead of tying the substrate to ground, the substrate is held at some negative voltage. Typically, this negative voltage is between -0.5 and -1 V. “Pumping” the substrate negative is found in some DRAMs. A negative substrate bias has several benefits. It (1) stabilizes n-channel threshold voltages, (2) increases latch-up immunity after power up, (3) prevents forward biasing n+ to p-substrate pn junction, (4) allows chip inputs to go negative without forward biasing a pn junction, (5) prevents substrate from going locally above ground, (6) reduces depletion capacitances associated with the n+ to p-substrate junction, and (7) reduces subthreshold leakage current.

A simple substrate pump is shown in Fig. 18.41. In this circuit we can use n-channel MOSFETs to generate a negative potential, unlike Fig. 18.35, since the negative voltage is connected to the substrate. In this situation, the drain/source implants of the n-channel MOSFETs cannot become forward biased. Note the absence of the load capacitance. It turns out that the capacitance of the substrate to everything else in the circuit presents a very large capacitance to the pump. In other words, the substrate itself is a very large capacitor.

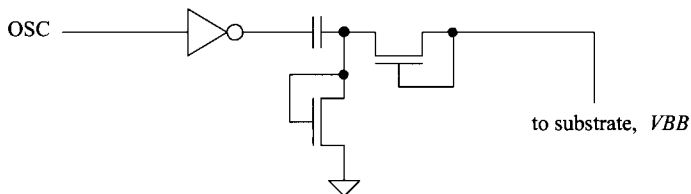


Figure 18.41 Simple substrate pump.

An example oscillator circuit that is used to drive the substrate pump is shown in Fig. 18.42. This is a standard ring oscillator with a NAND gate to enable/disable the oscillator. Capacitors are added at a few points in the middle of the oscillator to increase the delay and lower the frequency of the oscillator so that the pump's capacitors fully charge.

The final component in a substrate pump is the regulator, a circuit that senses the voltage on the substrate and enables or disables the substrate pump (the oscillator). Using

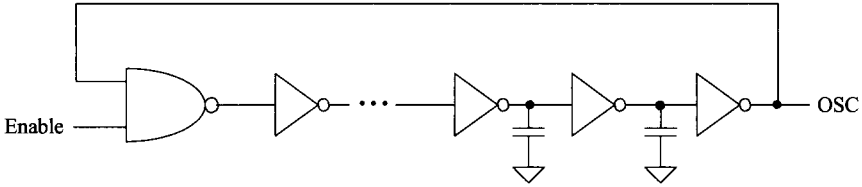


Figure 18.42 Ring oscillator with enable.

a comparator with hysteresis, a precision voltage reference, and a level shifting circuit, the enable signal can be generated with the circuit of Fig. 18.43. The hysteresis of the comparator determines the amount of ripple on the substrate voltage. Forcing a constant current through the two MOSFETs, M1 and M2, compels their source-gate voltages to remain constant (note how the body effect is eliminated by using p-channel MOSFETs), causing the MOSFETs to behave as if they were batteries. The battery action of M1 and M2 shifts the substrate voltage up so that it lies in the input range of the comparator. The number of MOSFETs (in this case two) and the magnitude of the current I determine the substrate voltage.

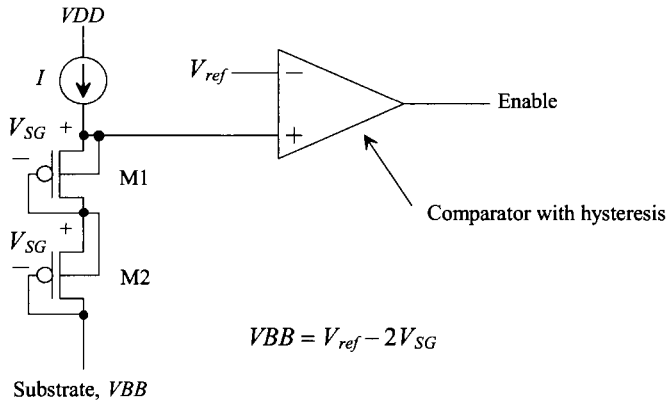


Figure 18.43 Regulator circuit used in substrate pump.

A simpler and less accurate implementation of the regulator is shown in Fig. 18.44. The comparator and voltage reference are implemented with an inverter and M1–M3. MOSFET M3 causes the inverter to have hysteresis, that is, behave like a Schmitt trigger. MOSFETs M1 and M2 form an inverter with a switching point voltage of approximately V_{THN} . The current source of Fig. 18.43 is implemented with the long L MOSFET M4 in Fig. 18.44. The level shifting is performed with the n-channel MOSFETs M5 and M6. When point A gets above a potential of V_{THN} , the Enable output goes high, causing the substrate pump to turn on and drive the substrate voltage negative. When point A gets pulled, via the substrate voltage through M5 and M6, below V_{THN} , the Enable output goes low and the pump shuts off. The substrate voltage generated with this circuit is approximately $-V_{THN}$ with body effect.